

Utilizing a Multi-Class CNN Model for Precise Diagnosis of Alzheimer's Disease

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Abstract—Early and advanced healthcare detection is crucial for identifying, diagnosing, and managing various health conditions. Alzheimer's disease(AD) is the leading cause of neurodegeneration, and initial stage detection of AD can reduce disease progression, improve prognosis, and increase survival rates with proper treatment. Magnetic Resonance Imaging (MRI) is an essential tool in diagnosing and monitoring AD, which helps visualize structural changes in the brain associated with the disease, such as shrinkage in specific regions like white matter, gray matter, cortex, and hippocampus, and enlarging the ventricles (fluid-filled spaces). Our main goal is to detect four stages of AD, namely Non-Demented(ND), Very Mild Dementia(VmD), Mild Dementia(MD), and Moderate Dementia(MoD). To classify these four stages, a new deep learning-based proactive 4-layer less complex 2D CNN model is built, and the dataset acquired from the Open Access Series of Imaging Studies-1 (OASIS-1) is fed to the model. SMOTE is applied to balance the dataset by generating synthetic samples for underrepresented classes for better accuracy. Unlike traditional models, our method enhances classification accuracy while maintaining computational efficiency. The proposed model gave a training and validation accuracy of 99.29% and 99.19%, respectively. It was also compared with ResNet50 and outperformed the state-of-the-art model on the same dataset. This lightweight model can be used in real-time AD detection and precise diagnosis applications.

Index Terms—MRI, Alzheimer's diseases, OASIS-1, SMOTE, CNN, Resnet50, Deep learning

I. INTRODUCTION

The human brain is an exceptionally influential and sophisticated organ, playing a central role in crucial functions like generating basic ideas, solving complex problems, thinking capabilities, making important decisions, imagining, and retaining memories. Memory is essential for storing and recalling experiences, allowing us to connect past experiences with present selves and anticipate the future. Memory loss due to dementia as well as Alzheimer's disease(AD) is a progressive decay in memory and cognitive function caused by damage to brain cells. Alzheimer's affects parts of the brain like the frontal lobes of the brain, entorhinal cortex, and hippocampus which are involved in memory functioning, storage, and retrieval, leading to increasing difficulty in memory functioning.

According to the Global Action Plan, over 55 million people worldwide are living with dementia [1]. Some recent analyses calculate that around 9.9 million people develop dementia globally every year; equating to a new case every

three seconds [2]. This is one of the less talked about human abnormalities, making this even deadlier. However, diagnosing AD manually is challenging because its symptoms overlap with normal aging and vascular dementia. Also, early detection of Alzheimer's disease helps slow progression, enables better treatment planning, and improves quality of life. It reduces long-term healthcare costs and supports caregivers while advancing research for potential treatments. Magnetic Resonance Imaging (MRI), Computer Tomography (CT) scans, functional MRI (fMRI), Electroencephalography (EEG), and Positron Emission Tomography (PET) are essential diagnostic tools for AD, enabling healthcare professionals to observe structural changes in the brain.

In EEG, specific patterns changing in brain waves with neurodegeneration are analyzed and extracts feature such as reduced power and phase locking in the beta and alpha bands, lower entropy, and reduction in Lempel-Ziv Complexity (LZC). Some of its limitations are lower accuracy, lacking precise information about deeper brain structures (e.g., hippocampus), inability to detect early stage, and the complexity of data interpretation [3]. When doctors diagnose a patient with AD, EEG alone isn't sufficient enough to confirm any anomalies. As a result, doctors additionally have to conduct MRI, PET scans, and cognitive assessments to reach any decision.

Another technique for diagnosing Alzheimer's is PET. The main features extracted from PET are the identification and quantification of amyloid plaque deposition in some brain regions, the decrease in glucose metabolism, and the reduction of hypometabolism in the precuneus, the posterior cingulate cortex, the temporal lobes, and the parietal lobes. However, PET has some drawbacks, such as high cost and limited accessibility, radiation exposure, limited sensitivity in early stages, complex interpretation, and the potential for false positives.

Due to the limitations of the procedures mentioned above, MRI scans are preferred over them nowadays. MRI scans can reveal shrinkage in brain regions associated with memory and reasoning abilities, like white and grey matter, the hippocampus, the temporal and parietal lobes, and other cortical areas, in the early stage. MRI can differentiate Alzheimer's from other types of dementia, monitor disease progression, and support early diagnosis. Moreover, MRI scans are noninvasive, widely

available, and radiation-free, making them suitable for our choice of dataset.

In early studies for the identification of AD, various machine learning methods were applied to structural MRIs like Tree Bagger(TB), K-Nearest Neighbors (KNN), Logistic Regression, and Support Vector Machine (SVM). In these feature-based methods, experts must define specific regions of interest (ROIs) in the brain, extract relevant features like standard deviation, mean, median, entropy, pixel value, etc. from these ROIs, and build a classification model [4]. However, this method heavily relies on intricate image pre-processing, which is both time-intensive and requires substantial computational resources to extract appropriate features. [5].

A major problem with this AD detection is the scarcity and imbalance of the dataset. For small-size data, we can adapt augmentation processes like rotation, flip, cropping, zoom, etc. SMOTE is another technique to handle the same. We have chosen SMOTE as it creates synthetic data and matches all classes of datasets. Data scarcity and class imbalance were also handled with a transfer learning approach, ResNet50 which was trained with the ImageNet dataset. In our work, we applied transfer learning to see and compare its metrics with our model. However, transfer learning requires a huge number of hyperparameters, which enhances the complexity of the model in terms of time and cost.

Our current work focuses on a customized CNN designed to improve the recognition and categorization of various stages of AD, from Non-Demented to Moderate Dementia. Loss of neurons and brain tissue leads to a noticeable shrinking of the brain over time which has to be detected earlier. Our current work focuses on a customized Convolutional Neural Network (CNN) designed to reduce the model's complexity, minimize computational requirements, and enhance the precision of AD detection. To identify the efficacy of the proposed framework, we have analyzed the model's metrics with and without SMOTE. Our proposed framework demonstrates remarkable performance compared to current leading techniques across various domains.

II. RELATED WORK

In the past decade, AD classification and detection have seen many approaches, which we can divide into three different clusters: feature extraction-based detection, transfer learning-based detection, and modified CNN-based detection. The researchers adopt Different types of approaches for this AD detection on a targeted basis from time to time.

Feature-based detection methods are widely used for diagnosing AD. In this category one important work comprised five steps [9]. First, Gray Matter (GM), White Matter (WM), and cerebrospinal fluid (CSF) are segmented by the voxel-based morphometric toolbox method. Then MRI data along with the Functional Assessment Questionnaire (FAQ) is multiplexed in the final stage to improve the classification accuracy. Likewise, Dinu A J et al. [16] proposed feature extraction methods using combined point detection. Random Forest and Tree Bagger classifier is used and feature was extracted manually which is

a big limitation in this process. That's why deep learning is practiced over the machine learning study.

To overcome these limitations, transfer learning(TL) based CNN models were introduced. In TL-based CNN methods some research groups test some pre-trained models. Shamrat et al. [7] tested TL architectures such as MobileNetV2, VGG16, ResNet50, InceptionV3, and AlexNet, and achieved better accuracy with Inception V3. Later they modified their model to design the AlzheimerNet using the MRI scan dataset acquired from ADNI. This 6-layer classification model was built based on Inception V3. Shruti et al. [6] performed a pre-trained CNN model, EfficientNetB0, to identify AD on the Kaggle dataset. Another popular classification model VGG 16 was used by Deepa et al. [11] for three different datasets; ADNI, OASIS, and SIMON databases. CAT12 toolbox is used in this research to process T1-weighted MRI images to balance the uneven light in MRI images and this research gives a better result.

One more approach for AD detection is the CNN model with modified layer and hyperparameters and this approach has seen more success in terms of AD detection. Odusami et al. [10] presented a modified ResNet18 model to identify the initial stage of AD where he has extracted the features from neuroimaging data and the data set is gathered from ADNI. Mehmood et al. [12] processed FMRI images collected from ADNI and fed them into a multi-layered CNN model built by themselves consisting of a modified convolution layer, pooling layer, and activation layer. In [8], Zhang et al. used two CNN models on the ADNI dataset, and the correlation analysis measured the final output. for the diagnosis of AD. This network incorporated a convolutional autoencoder for unsupervised feature learning and a softmax regression layer for classification. Fathi S et al. [13] proposed a combination of 6 different CNN models which gave a better accuracy rate than a single CNN model. Santosh Kumar Tripathy et al. [14] proposed an innovative approach that involves the development of an improved Spatial Attention Block (I-SAB) designed to create spatial attention maps within multilayer CNN. Experimentation is done using OASIS and AD-Dataset. The highest success rate of this section motivated us to work with a modified CNN model for AD detection.

III. METHODOLOGY

Our goal is to propose an end-to-end framework to identify different stages of AD precisely and indicate the required treatment level.

A. Dataset Details

We have used the OASIS-1 dataset for our training [17]. This dataset consists of T-1 weighted MRI scans which are collected from 416 subjects aged from 18 to 96 both male and female. Among the 3 types of MRI which are Axial, Coronal, and Sagittal, axial MRI is considered for this study. The brain images were sliced along the z-axis into 256 pieces and slices ranging from 100 to 160 were selected from each patient. The dataset is classified into four labels; ND, VmD, MD, and MoD which is done based on Clinical Dementia

Rating (CDR) values. There are 67222 images in the ND dataset, 13725 images in VmD, 5002 images in MD, and 488 images in the MoD classes. Due to limitations in Hardware capacity, we took 5000 images from the first three labels and 488 images from the MD label.

B. Data Preprocessing and Feature Preparation

The framework consists of five main layers namely, data acquisition, data preprocessing, data splitting for training and testing, building the customized CNN model, and AD classification as shown in Fig. 1.

In data acquisition, images of 496×248 pixels are taken as input in different slices of axial view MRI. After the filtration process such as histogram matching, smoothening, and intensity adjustment, images are resized to 128×128 for further processing. Then OneHotEncoding is applied to convert labels into categorical data. Due to the imbalanced dataset, SMOTE is performed which generates synthetic examples rather than just duplicating existing minority class examples. SMOTE was applied to this dataset to equalize the number of images for each class. After SMOTE each class had 5000 images randomly selected from OASIS-1. Later We divided the data for training and testing in a ratio of 80% and 20% respectively and further divided the training data into training and validation sets comprising 90% and 10% respectively and the training portion is fed into the customized model. Finally, results were compared among various combinations of models and data.

C. Classification Model

1) *Customized CNN model*: The proposed CNN model consists of various layers. Four 2D-convolution layers constitute

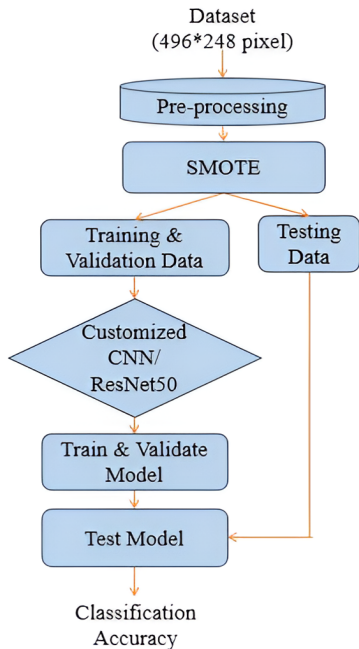


Fig. 1. Overview of the proposed work.

the proposed model, shown in Fig. 2, each having a kernel size of (2×2) and filter sizes varying from 16 to 512. Batch normalization and pooling layers have been added after every convolution layer. In addition, a dropout layer of 0.25% has been used to prevent overfitting.

Activation function ReLU is used after every convolution layer and SoftMax is used for the output dense layer as this is a multiclass classification problem we are dealing with. Adam optimizer was used which is suitable for most deep learning tasks and can handle large data efficiently. We started with a learning rate of 0.001 which is efficient for our model. The minimum learning rate was declared as 0.00001 with a batch size of 90. In the Callback section, we used Earlystopping and Learning rate reducer which used validation accuracy to decide where to stop learning.

2) *ResNet50*: He et al. [15] proposed one of the efficient deep learning-based CNN models named ResNet which we used in our work. The usage of shortcut connections is the specialty of this model. X in the term ResNet-X model means Residual deep neural network with X number of layers. ResNet-50, and ResNet-101 are some of the popular models in this series. To evaluate the effectiveness of our proposed model we have performed a pre-trained model ResNet-50 on our dataset. We simply called the ResNet-50 model directly and used the same set of hyperparameters and callback functions as earlier.

D. Experimental Setup

Implementing a neural network requires a good setup of hardware for smooth training of the model. The operating system we used was Windows 11 Pro, OS build 22631.4317. This operating system was installed on a Central Processing Unit (CPU) having a 12th-generation Core i5-1235U processor. The operating system was 64 bit variant and the installed RAM was 16 GB. The processor speed was 1.30 GHz with a top speed of 1.80 GHz.

For running our program which was written using Python language was implemented on Jupyter Notebook in Anaconda Navigator having an environment with TensorFlow (v2.10.0) background and Keras as part of the TensorFlow (v2.10.0) library. This customized package allowed us for a flexible,

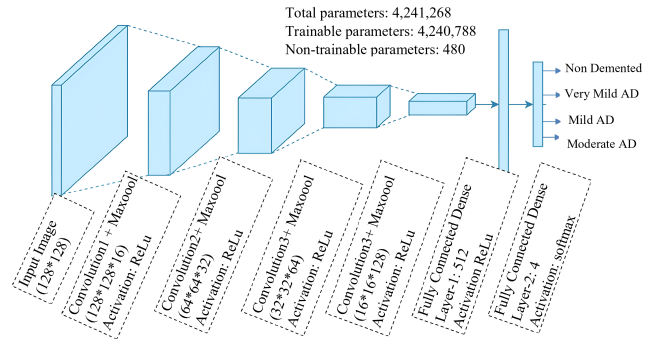


Fig. 2. Model architecture of customized CNN model.

controlled workspace for deep learning model development with lots of updated libraries.

IV. RESULTS AND DISCUSSION

This section describes the outcomes obtained by our customized framework with SMOTE and without SMOTE and the transfer learning model in terms of various metrics of neural networks.

A. Customized CNN without SMOTE

We trained our customized model with an imbalanced dataset to observe its performance. Due to the earlystopping function training stopped after 34 epochs and we got 99.39% training accuracy and 92.86% validation accuracy as shown in Fig. 3. Model training and validation losses were 1.76% and 22.77% respectively which are shown in Fig. 4. From the curves, it can be seen that getting a stabilized accuracy and loss is difficult in this case.

B. ResNet50 without SMOTE dataset

ResNet 50 is a pre-trained model, trained on larger datasets like ImageNet. This model was implemented on the same dataset, epochs, batch size, and learning rate. The result we found can be seen in Fig. 5 and Fig. 6 having a training accuracy, and validation accuracy, of 97.81%, and 96.34% respectively.

C. ResNet50 with SMOTE dataset

After applying SMOTE, the balanced dataset is fed to the ResNet model. We found a training and validation accuracy of 99.75% and 97.50% which can be seen in Fig. 7 and Fig. 8. This model is a complex one with 25,689,988 hyperparameters which requires high computational power and time taking 141 seconds per epoch approximately.

D. Customized CNN with SMOTE

We simplified the deep learning-based CNN model and applied the model to the OASIS dataset which was balanced using SMOTE. A pre-processed dataset was trained and gave us a more satisfactory result. We obtained 99.91% training accuracy and 99.67% validation accuracy, which indicates

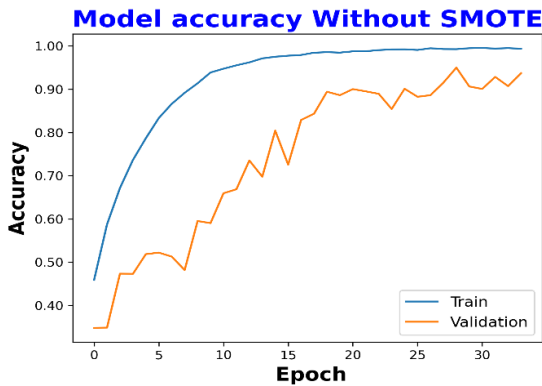


Fig. 3. Accuracy curve of the customized CNN without SMOTE.

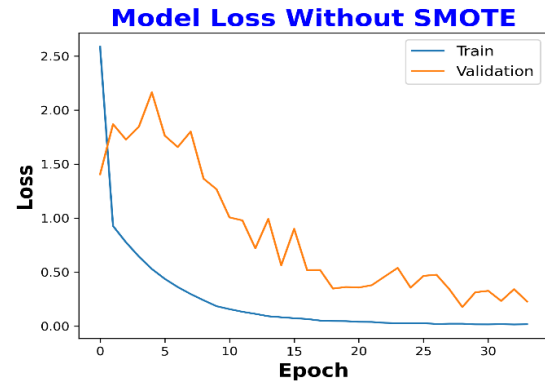


Fig. 4. Loss curve of the customized CNN without SMOTE.

better performance than the other instances. Accuracy and loss are shown in Fig. 9 and Fig. 10. This model requires only 4,256,488 hyperparameters and 73 seconds per epoch. We have also plotted the confusion matrix for our model shown in Fig. 11. The pre-trained model performs satisfactorily with limited data but as we have large data and SMOTE is applied to balance labels, our customized model has been proven to

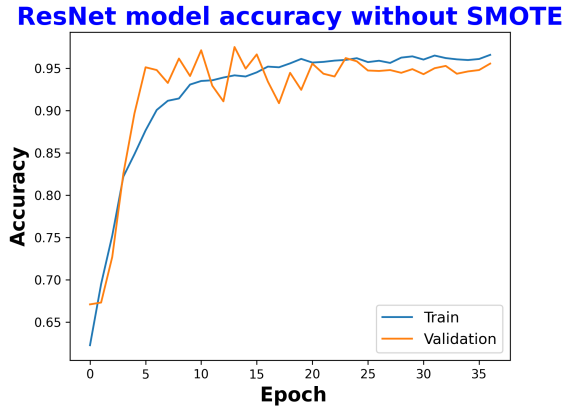


Fig. 5. Accuracy curve of ResNet50 without SMOTE.

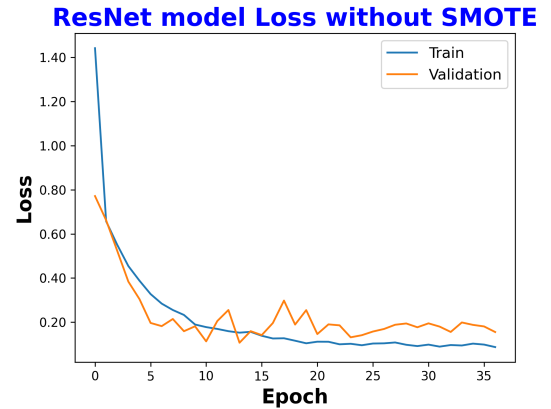


Fig. 6. Loss curve of ResNet50 without SMOTE.

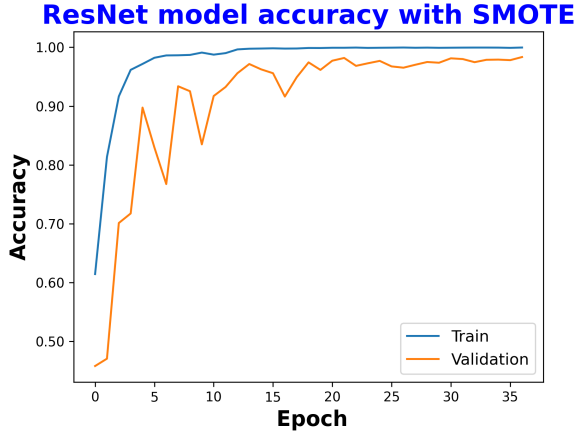


Fig. 7. Accuracy curve of ResNet50 with SMOTE.

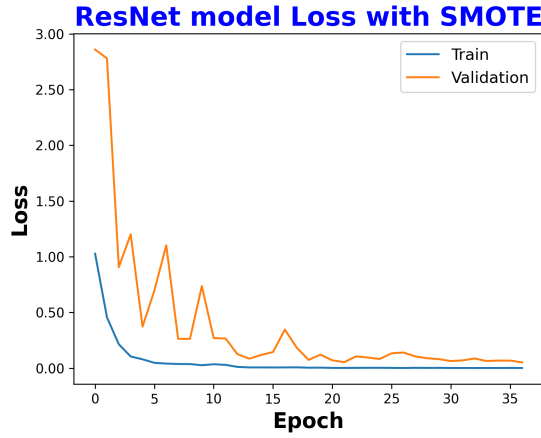


Fig. 8. Loss curve of ResNet50 with SMOTE.

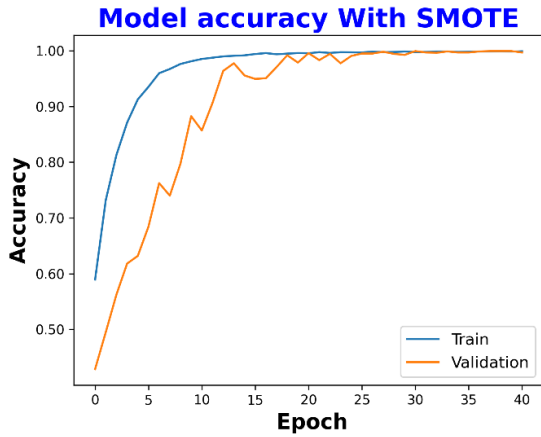


Fig. 9. Accuracy curve of the proposed framework with SMOTE.

be simple and cost-effective as from the above discussion.

E. Metrics

Metrics used for comparison among our tested models were Test Accuracy, Recall, Precision, and F-1 Score. Table I it is

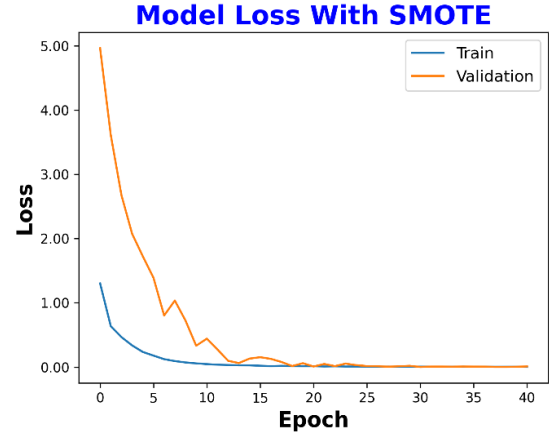


Fig. 10. Loss curve of the proposed framework with SMOTE.

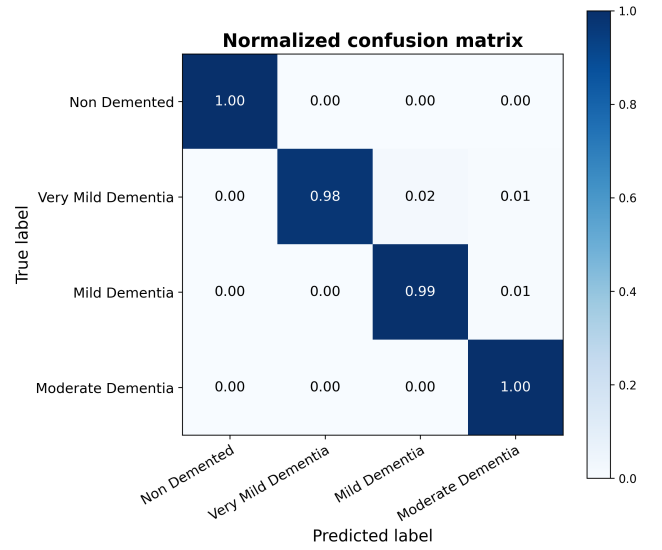


Fig. 11. Normalized confusion matrix of the proposed framework with SMOTE.

self-explanatory that our proposed customized model had a better outcome compared to the other two models.

TABLE I
METRICS COMPARISON AMONG APPLIED APPROACHES

Models	Metrics Evaluation			
	Accuracy	Recall	Precision	F1-score
ResNet50 (Without SMOTE)	96.34%	96.08%	95.93%	95.67%
Proposed Model (Without SMOTE)	92.86%	91.98%	92.23%	91.07%
ResNet50 (With SMOTE)	97.50%	98.30%	98.54%	99.02%
Proposed Model (With SMOTE)	99.67%	99.96%	99.94%	99.92%

For more clarification on how well our model performed, we evaluated the model with some state-of-the-art models that were tested in various experimental setups by various

researchers. This evaluation can be seen in Table II. It is seen that our model Outperforms models like SVM et al. [9], Random Forest et al. [16], various pre-trained CNN architectures (e.g., ResNet50, InceptionV3) et al. [7] and modified CNN et al. [14] have an accuracy of 97.01%, 98.42%, 80.98%, 96.31%, and 99.75% respectively. While ResNet18 achieves 99.95% accuracy, it comes with higher computational costs due to its deeper architecture and numerous hyperparameters. In contrast, our Modified CNN with SMOTE (99.93%) offers nearly the same accuracy with a lighter and more efficient model. From the comparison, it is evident that our proposed framework is preeminent to traditional deep learning methods used by other researchers as this overcomes the data imbalance problem as well as the model complexity issue of transfer learning.

TABLE II
COMPARISON AMONG PROPOSED AND EXISTING MODELS

Reference	Modality	Used Model	Accuracy
Beheshti I et al. [9]	MRI, Functional Assessment, Questionnaire (FAQ)	SVM	84.07%, 97.01%
Dinu A J et al. [16]	MRI	Random Forest, Tree Bagger	98.42%, 98.17%
Shamrat et al. [7]	MRI	VGG16, MobileNetV2, AlexNet, ResNet50, InceptionV3	78.84%, 86.85%, 78.87%, 80.98%, 96.31%
Shruti et al. [6]	MRI	EfficientNetB0	95.78%
Odusami et al. [10]	fMRI	ResNet18	99.95%
Santosh Kumar Tripathy et al. [14]	MRI	CNN	99.75%, 96.20%
Proposed Model with SMOTE	MRI	Modified CNN	99.93%

V. CONCLUSION

This paper presents a deep learning-based customized 2D CNN model using a T-1 weighted MRI image dataset to detect AD and its four stages automatically. Firstly, we used our customized model on the OASIS-1 dataset which was an unbalanced dataset. Further, we performed SMOTE on the same dataset to solve the data imbalance issue and again applied the customized model to it. On a balanced state, our model performed better achieving 99.91% and 99.67% training and validation accuracy respectively. Later, we used ResNet50 on the balanced dataset and recorded the metrics values. It was found that our customized CNN model outperformed the state-of-the-art model ResNet50 in terms of accuracy. Other metrics such as Recall, Precision, and F1-Score were also very satisfactory with a higher value than the other models. As this proposed model is simple, requires fewer hyperparameters, has higher accuracy, and stable loss curve, this model is well-suited for AD detection. As this model is simple in structure, it

saves computational costs also. In the future, we will work on diverse datasets and explore hybrid models for better accuracy and precision in AD detection.

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